



MINING DEALER BEST PRACTICE

Machine Inspection Toolbox

**Maintenance and
Repair**

Application

Component Life
Management

Component
Renewal (CRC)

MARC
Management

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April 2021
1.10-210413-1

1.0 Introduction

For machine inspections, the general tool requirements are:

- Pressure test gages
- Pressure hoses
- Blowby hose
- Blowby meter and its attachments
- Computer
- ET control device
- Aerometer
- Thermometer
- Hydrometer battery tester
- Voltmeter
- Filter cutter tool

This best practice provides an improvement in two areas:

1. Handling

Handling this equipment in separate individual carry cases can or result six or seven cases being used. Since the cases can be heavy, the technician would need to make several trips (time consuming) or risk damaging boxes and equipment. Those plastic cases are often too firm, intending to protect the attachments and devices, but still often becoming damaged as they move around.

2. Safety

When attempting to carrying out a variety of pressure tests with a limited number of gages available, a safety risk can occur. The safety risk occurs when a hoses and gages are required to be handled by the technician (both removal and re-fitting) in order to test various pressures without having sufficient time to allow oil and components to cool.



Figure 1 - Broken Plastic Case

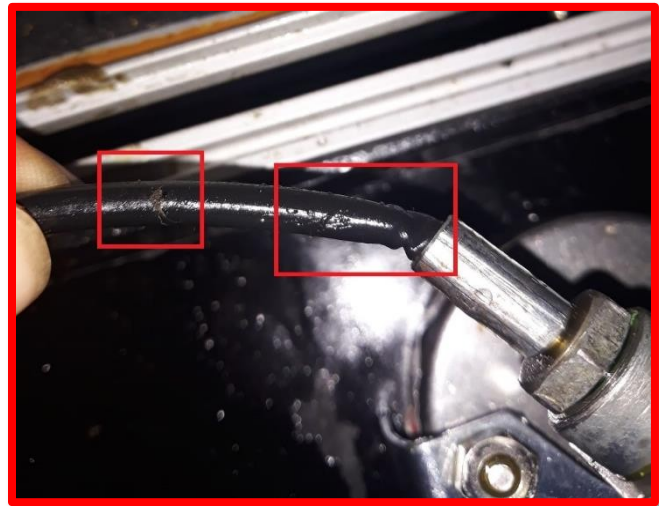


Figure 2 - Broken Hose

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Figure 3 - Old Plastic Cases

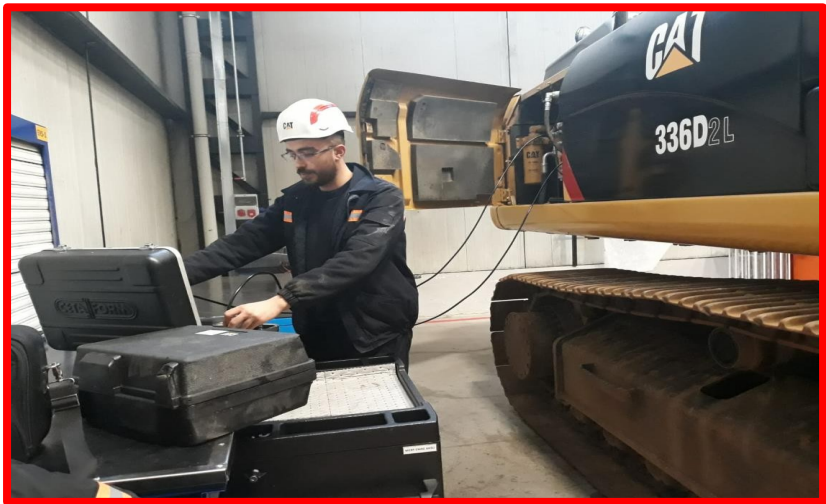


Figure 4 - Old Inspection Method

This Best Practice solution is to provide a single ‘Machine Inspection Technician Toolbox’.

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2.0 Best Practice Description

The Machine Inspection Technician Toolbox enables its user to carry its inspection devices and attachments easily, quickly and safely.

The toolbox consists of 5 shelves containing the following:

1. Twenty pressure test gages of various ratings
2. Aerometer, Thermometer, Hydrometer battery tester, Voltmeter and a Filter cutter tool
3. Computer and ET control device
4. Blowby meter and attachments
5. Pressure hoses and a blowby hose

The fifth shelf is larger than other shelves providing space for longer hoses. Styrofoam is used to secure the equipment. More details and a sketch can be found in the Supporting Attachments / References part.

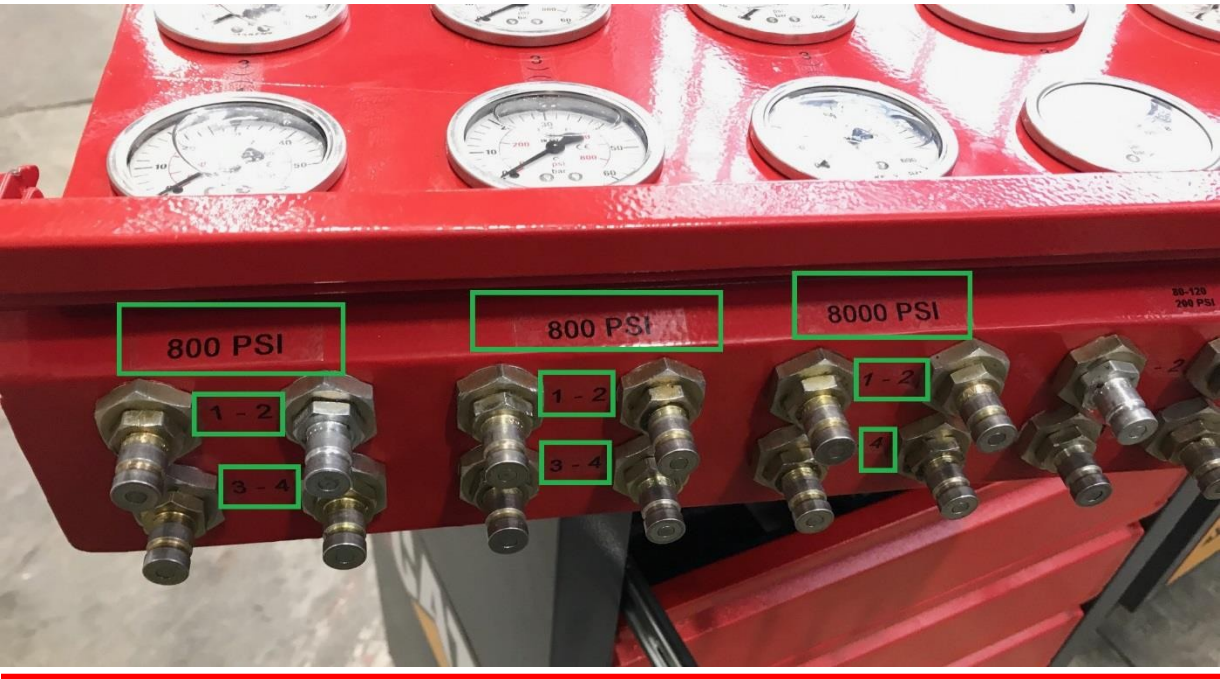
3.0 Implementation Steps

Step1: Connecting pressure test gages to metal plate



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Step2: Labeling pressure test gages according to their order and pressure value



Step3: Shaping styrofoam for placing devices and attachments



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Step4: Placing devices and attachments



Step5: Placing devices and its attachments to the Inspection Box



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4.0 Benefits

1. Reduces inspection time from 1 days to around 2 hours
2. Reduces number of journeys to and from equipment store carrying multiple plastic cases
3. Prevents occupational-work accidents which are based on lack of equipment.
4. Increases quality of pressure measurement.

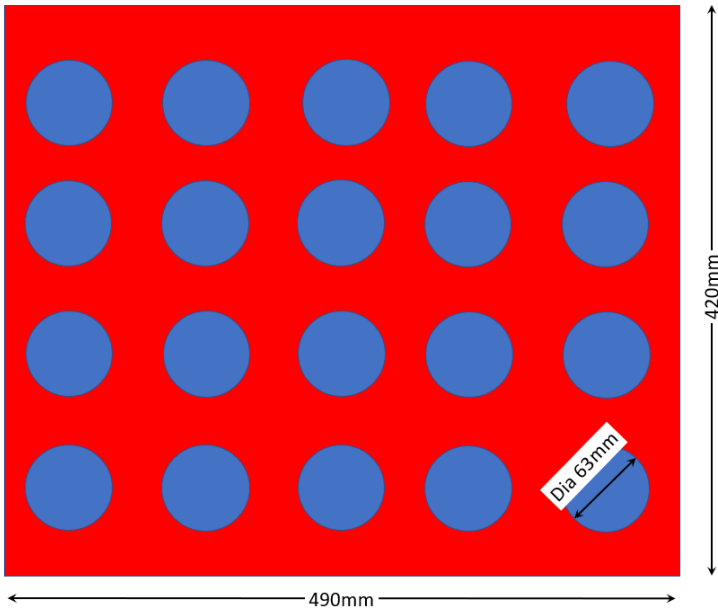
5.0 Resources Required

Item	Cost
Pressure Test Gages(20 pieces)	\$160
Houses and Coupling	\$400
Total	\$560

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6.0 Supporting Attachments / References

A sketch of the metal sheet where pressure test gages placed in.



7.0 Related Best Practices

None.

8.0 Acknowledgements

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